Alexa Business Autopilot

Hackathon Strategy: Voice-first operating layer for Indian small businesses

Working thesis: Instead of building “Amazon Now for businesses,” build an Alexa-powered business agent that understands demand, inventory, cash constraints, capacity and customer opportunities, then recommends or executes the next best action. Amazon Now is one possible supply-side actuator.

1. Executive Summary

The opportunity is to make Alexa+ useful to owner-operated Indian businesses that often run on WhatsApp, phone calls, notebooks and fragmented apps. The agent should let an owner speak naturally about the state of the business and receive an actionable recommendation.

- Core promise: “Tell Alexa what is happening in your business; Alexa figures out what should happen next.”
- Primary use case: cash-aware, urgency-aware procurement and replenishment.
- Expansion: capacity-to-customer matching, business memory, document reconciliation and customer acquisition.
- Design principle: voice-first, but multimodal when a screen is available. Alexa+ can render voice responses, inline UI and fullscreen UI; headless devices must still receive a complete voice experience.
- Important constraint: current Alexa+ Category SDK and MCP Toolkit documentation says these tools are available to select partners; hackathon access must be verified before implementation.

2. Target Business Archetypes

Business	Procurement	Customer acquisition	Why useful	Priority
Kirana / mini grocery	Very high	Medium	Maintaining payment records	High
Tiffin / cloud kitchen	Very high	Very high	Demand uncertainty + ingredients + packaging+ order tracking	Very high
Bakery	Very high	High	Perishables + demand forecasting	Very high
Salon / beauty	Low-Medium	Very high	Appointments, repeat customers, unused capacity	High
Tailor / boutique	High	High	Materials + custom orders	High
Laundry / dry cleaner	Medium	Very high	Pickup, processing, delivery and repeat demand	High
Repair technician	High	Very high	Parts + scheduling + local leads	Very high
Café / small restaurant	Very high	Very high	Inventory + demand + reservations/orders	Very high

Carpenter / furniture	High	High	Raw material + custom orders	Medium
Home-service professional	Medium-High	Very high	Local discovery + scheduling + supplies	Very high

3. Common Value Chain

Across these businesses, the recurring loop is: Demand → Operations → Inventory/Capacity → Procurement → Fulfillment → Payment → Retention, with Finance surrounding the loop.

The shared pain is not simply “find a product.” It is: How much should I buy? When should I buy it? Can I afford it? What happens if I wait? Do I have capacity to take another customer? How do I get that customer?

4. Proposed Product: Alexa Business Autopilot

Core conversational example:

Owner: “Alexa, we have 20 tiffin orders for tomorrow. Do we have enough ingredients and can I afford what is missing?”

Agent flow: inspect inventory → estimate demand → calculate missing quantities → check cash/budget → compare urgent vs normal procurement → recommend the lowest-risk action → request approval → create/execute purchase where authorized.

The agent should expose business tools such as:

- get_inventory()
- get_cash_position()
- get_upcoming_payments()
- forecast_demand()
- calculate_reorder_quantity()
- optimize_purchase()
- analyze_document()
- get_capacity()
- find_customer_opportunities()
- create_purchase_plan()

5. High-Potential Novel Features

Idea	What it does	Novelty angle
Cash-aware instant procurement	Decide whether to buy now, buy partially, or wait. Model purchase cost + delivery premium + expected stockout cost + cash constraints.	Potentially differentiated when combined into a single decision loop; validate with deeper prior-art search.
Capacity-to-customer matching	A business tells Alexa when it has unused slots/capacity. The system can expose that capacity to relevant customer-side discovery flows where integrations permit.	Potentially differentiated when combined into a single decision loop; validate with deeper prior-art search.

Business memory	Remember structured business facts such as supplier prices, recurring demand, past delivery problems and owner preferences, while distinguishing verified records from informal statements.	Potentially differentiated when combined into a single decision loop; validate with deeper prior-art search.
Document-aware reconciliation	Extract invoice/order information and detect quantity, price or tax discrepancies before the transaction is recorded or paid.	Potentially differentiated when combined into a single decision loop; validate with deeper prior-art search.
Voice-first Bharat business OS	Let owners operate the system in natural English/Hindi/Hinglish instead of learning ERP terminology or navigating multiple dashboards.	Potentially differentiated when combined into a single decision loop; validate with deeper prior-art search.

6. Current Alexa+ Capabilities to Build Around

Capability	Status	Relevance
Natural conversation	Available	Alexa+ supports multi-turn natural conversation and follow-ups.
English / Hindi / Hinglish	Available in India	Amazon states Alexa+ launches in India with English, Hindi and Hinglish.
Memory / personalization	Available	Alexa+ can remember preferences and use context.
Document understanding	Available	Users can upload documents/images and ask Alexa+ questions about them.
Shopping	Available	Product discovery, comparisons, review summaries, shopping lists and purchasing assistance.
Amazon Now	Announced / coming in India	Amazon says groceries and daily essentials can be ordered through Alexa+ via Amazon Now.
Local booking	Alexa+ category	Supports search, availability, appointment creation, rescheduling and cancellation for supported local businesses.
Home services	Alexa+ category	Supports service discovery/booking flows for home services.
Food ordering	Alexa+ category	Supported by Category SDK.
Restaurant reservations	Alexa+ category	Supported by Category SDK.
Ticketing	Alexa+ category	Supported by Category SDK.
Ride booking	Alexa+ category	Supported through Alexa+ category integration.
Custom MCP add-ons	Developer capability, partner-gated	Can expose custom tools; Alexa+ can orchestrate them through natural language.
Interactive UI	Supported for MCP Apps	Screen-capable devices can show inline/fullscreen experiences; headless devices require voice-complete behavior.
Checkout	Supported for eligible add-ons	Alexa+ can manage confirmation/payment flow; backend

		provides commerce endpoints when using the supported integration.

7. Existing Amazon / Alexa Solutions We Must Not Duplicate

- Amazon Seller Assistant: already handles listings, inventory recommendations, business insights, pricing-related guidance and global expansion for Amazon sellers, including conversational English/Hinglish.
- Amazon Business: business purchasing, spend controls, organizational purchasing and related workflows already exist.
- Alexa+ shopping: already handles consumer product discovery, comparisons, shopping lists and purchasing.
- Alexa+ local booking / home services / food ordering / restaurant reservations: already provide customer-side discovery and transaction flows in supported categories.
- Therefore, “AI assistant + inventory” or “voice shopping for businesses” alone is not a strong novelty claim.

8. The White Space We Should Own

The strongest positioning is the business-side decision loop: Demand → Business state → Financial constraint → Procurement/capacity decision → Action. Amazon Now should be an actuator, not the entire product.

Potential research question: Can a voice-first agent jointly reason over demand, inventory, cash constraints, capacity and rapid-delivery options to reduce stockouts and unnecessary procurement for owner-operated small businesses?

9. Concrete Hackathon MVP

Phase	Deliverable
Choose one archetype	Use a tiffin/cloud-kitchen or café because it demonstrates both procurement and customer demand.
Create synthetic business state	Inventory, daily demand, upcoming orders, cash balance, supplier ETA, Amazon Now-like catalog.
Build 6–8 tools	Inventory, forecast, cash, upcoming payments, procurement optimizer, basket builder, document checker, capacity.
Build Alexa integration	Use the hackathon-provided Alexa+ path if available. Otherwise create the closest permitted Alexa prototype and keep the business server MCP-ready.
Implement voice-first flow	Every critical task must work without a screen.
Add optional visual dashboard	On screen-capable Alexa+ surfaces, show stock, recommended basket, cost and rationale.
Add one document workflow	Upload an invoice; extract values; compare against the proposed purchase and flag discrepancies.
Add one customer-acquisition flow	Owner declares free capacity; system creates a customer opportunity/recommendation using a simulated or authorized local-discovery connector.

10. Suggested 7-Day Build Plan

Day	Goal

Day 1	Lock target business + user journeys; verify hackathon Alexa+ access and allowed APIs.
Day 2	Build business-state backend and synthetic dataset.
Day 3	Implement procurement/cash optimization and business-memory layer.
Day 4	Connect Alexa integration; implement core voice dialogue and tool calls.
Day 5	Add document reconciliation + optional screen UI.
Day 6	Add customer-capacity flow, error handling and demo data; run evaluation.
Day 7	Polish UX, record fallback demo, prepare architecture/novelty slides and final pitch.

11. Demo Script

1. 1. Owner: “Alexa, we have 20 tiffin orders tomorrow. Check whether we have enough stock.”
2. 2. Alexa checks inventory and forecast.
3. 3. Alexa: “You are short on oil and containers. Buying everything now would cost ₹X.”
4. 4. Owner: “Can I afford it?”
5. 5. Alexa checks cash and upcoming obligations.
6. 6. Alexa recommends a constrained purchase and explains why.
7. 7. Owner: “Order what we need.”
8. 8. Alexa prepares the purchase/checkout through the permitted integration or a clearly labelled simulator.
9. 9. Show an invoice/document discrepancy and ask Alexa to flag it.
10. 10. Owner: “I have two free delivery slots tomorrow afternoon.” Alexa records capacity and demonstrates the customer-acquisition/discovery flow where supported.

12. Evaluation Metrics

- Procurement cost and emergency-delivery premium
- Stockouts avoided
- Cash-budget violations avoided
- Unnecessary inventory purchased
- Forecast/reorder recommendation accuracy
- Invoice discrepancy detection accuracy
- Number of conversational turns to complete a task
- Owner approval rate of recommendations
- Customer-capacity utilization / leads generated in the simulated flow
- Percentage of critical workflows that remain usable without a screen

13. Risks and Constraints

- Alexa+ Category SDK and MCP Toolkit are currently documented as select-partner capabilities. Verify hackathon access before committing to a production-style add-on.
- Amazon Now ordering should not be assumed to expose a public developer API. Use only authorized integrations; otherwise simulate the commerce backend and label it clearly.
- Document extraction is not the same as document authenticity verification. Do not claim a document is genuine unless an authoritative verification source confirms it.
- Financial recommendations should be explainable and approval-gated rather than fully autonomous.

- Privacy/security: business financial, customer and document data should be minimized, authenticated and scoped to the stated purpose.

14. Official References

Alexa+ Developer Docs: https://developer.amazon.com/docs/alexaplus/add-ons/home.html

Alexa+ for Builders: https://developer.amazon.com/alexaplus/

Alexa+ Category SDK Overview: https://www.developer.amazon.com/ja/docs/alexaplus/add-ons/overview-category-sdk.html

Alexa+ MCP Toolkit Overview: https://developer.amazon.com/zh/docs/alexaplus/add-ons/mcp-toolkit-overview.html

Local Booking SPI: https://www.developer.amazon.com/docs/alexaplus/add-ons/local-booking-spi.html

Alexa+ Checkout: https://developer.amazon.com/docs/alexaplus/add-ons/implement-checkout-endpoints.html

Alexa+ India: 25 Things to Try: https://www.aboutamazon.in/news/device-and-alexa/things-to-try-new-alexa-plus

Alexa+ India Launch: https://www.aboutamazon.in/news/devices-and-alexa/alexa-plus-launch-india

Amazon Seller Assistant India: https://www.aboutamazon.in/news/small-business/amazon-seller-assistant-ai-tool-india

Supplier communication

Payment tracking

Bulk ordering

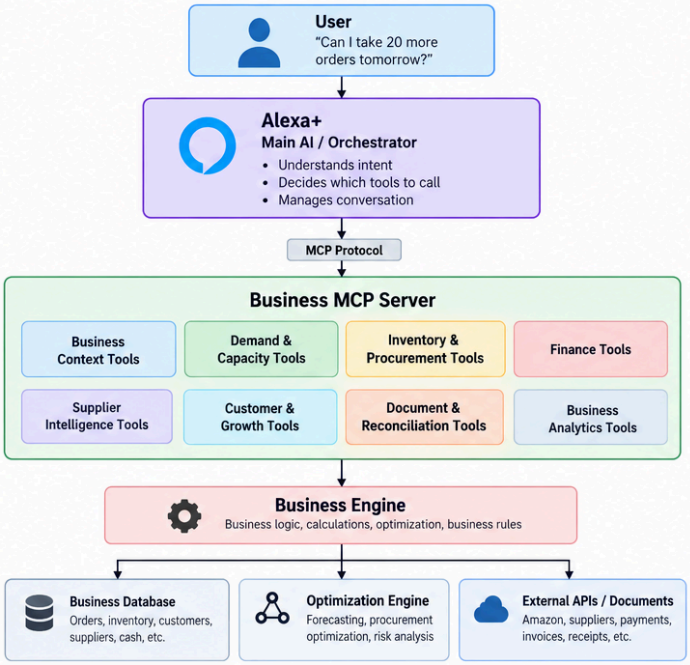
Order tracking

Inventory tracking

Local leads/Target customers outreach

Customer communication

1. Overall Architecture

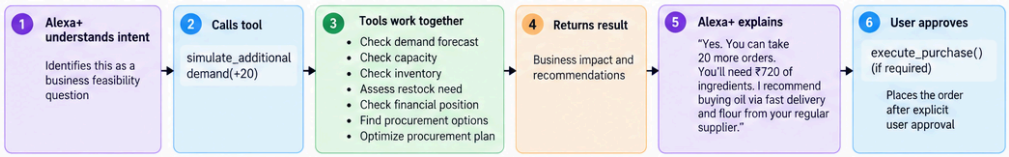


2. Agents and Tools

Agent	Purpose	Tools (Examples)
Business Context Agent	Maintains overall business state and recent changes.	• get_business_overview() • get_recent_business_changes() • get_business_memory()
Demand & Capacity Agent	Predicts demand and checks if business can handle it.	• forecast_demand() • analyze_demand_patterns() • get_capacity_status() • simulate_additional_demand()
Inventory & Procurement Agent	Determines what to buy, when, from whom and how much.	• analyze_inventory() • assess_restock_need() • find_procurement_options() • optimize_procurement_plan() • create_purchase_plan() • execute_purchase()
Finance Agent	Analyzes cash, profitability and financial feasibility.	• get_financial_position() • forecast_cash_flow() • analyze_profitability() • analyze_purchase_affordability() • simulate_financial_impact()
Supplier Intelligence Agent	Evaluates suppliers and maintains supplier relationships.	• get_supplier_profile() • compare_suppliers() • get_supplier_history() • evaluate_supplier() • update_supplier_record()
Customer & Growth Agent	Understands customers and identifies growth opportunities.	• analyze_customer_base() • analyze_customer_feedback() • analyze_customer_demographics() • identify_growth_opportunities() • simulate_customer_campaign()
Document & Reconciliation Agent	Extracts and verifies invoices/ receipts and detects discrepancies.	• extract_business_document() • reconcile_document() • detect_transaction_discrepancy() • record_verified_transaction()
Business Analytics Tools	Analyzes performance and generates reports.	• analyze_business_performance() • analyze_product_profitability() • analyze_cost_trends() • generate_business_report()

3. Example Tool Calling Flow

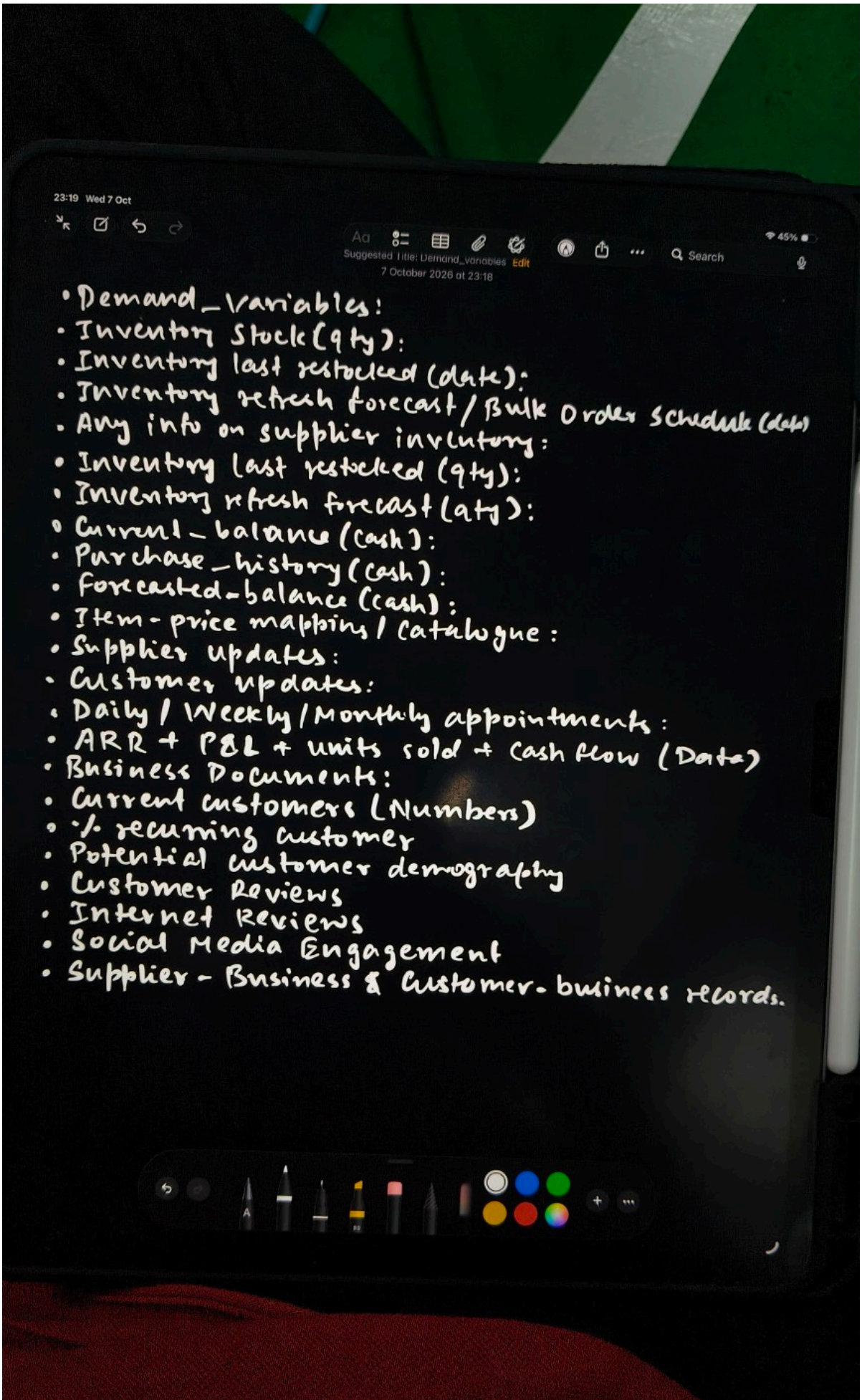
User: "Can I take 20 more orders tomorrow?"



4. Tool Types and Usage

Deterministic Tools (Calculation / Retrieval)
Used for database queries, calculations and business rules.
Examples: get_business_overview(), get_financial_position(), analyze_inventory(), forecast_demand(), get_supplier_history()

Specialist Agents (Complex Reasoning)
Used where multi-step reasoning, trade-offs or interpretation is required.
Examples: Inventory & Procurement Agent, Customer & Growth Agent, Document Agent.


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45%
• Demand_variables:
• Inventory Stock (qty):
• Inventory last restocked (date):
• Inventory refresh forecast / Bulk Order Schedule (date)
• Any info on supplier inventory:
• Inventory last restocked (qty):
• Inventory refresh forecast (qty):
• Current_balance (cash):
• Purchase_history (cash):
• Forecasted_balance (cash):
• Item-price mappins / Catalogue:
• Supplier updates:
• Customer updates:
• Daily / Weekly / Monthly appointments:
• ARR + P&L + units sold + Cash flow (Data)
• Business Documents:
• Current customers (Numbers)
• % recurring customer
• Potential customer demography
• Customer Reviews
• Internet Reviews
• Social Media Engagement
• Supplier - Business & Customer - business records.